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Gender, Group Composition, and Task Type in Small Task Groups Using Computer-Mediated Communication

Victor Savicki, Merle Kelley, and Dawn Lingenfelter

Western Oregon State College

Abstract — *In small (4–6 member), online task groups two factors were varied: (a) group composition, in terms of the gender of the group members, and (b) assigned tasks, in terms of the content and amount of cooperation required. Gender group composition included female only (FO), male only (MO), and evenly mixed male and female (MIX) groups. The two task conditions included a 'feminine'-content, decision-making or a 'masculine'-content, intellective task. Groups came to consensus on the task answer using only asynchronous computer-mediated communication (CMC). It was predicted that FO and MO groups would demonstrate communication and satisfaction differences as a function of task assigned as well as group composition. Group composition was related to many group process variables in significant ways; however, in general, task differences were less strong. FO groups, regardless of task, sent more words per message, were more satisfied with the group process, and reported higher levels of group development than either MIX or MO groups. However, both task and gender composition variables were related to various measures of choice of language. Mixed results with regard to gender composition and choice of language require a further examination of gender effects on CMC as occurring in*

Requests for reprints should be addressed to Victor Savicki, Psychology Department, Western Oregon State College, Monmouth, OR 97361, USA. E-mail: savickv@fsa.wosc.osshe.edu

small task groups. Choice of language relation to task type were generally opposite of predictions and require clarification of task distinctions and methodologies used. The significance of the results lies in defining the styles of communicating in the CMC context that will enhance group development.
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Recently, both the popular press (Tannen, 1994) and research studies (Herring, 1994) have attended to the increased use of networked computers and the relationship between gender and computer-mediated communication (CMC). Such examination is timely since more and more emphasis is being placed on CMC in education and in the workplace. Historically, CMC has been a male-dominated domain (Kirk, 1992). However, women are taking advantage of this mode of communication in ever growing numbers. Previous studies cite both situations in which CMC is a hindrance to full participation for both genders (Herring, 1994) and situations in which CMC enhances communication for both genders (Herschel, 1994). The current study, building on the results of previous research (Savicki, Kelley, & Lingenfelter, 1996) will isolate a specific variable, the type of group task, to determine whether or not it has a relationship to online communication or group variables. The following brief literature review will place the current study in a larger research context concerning gender and CMC.

The overall framework of this study, as in the previous study (Savicki et al., 1996), focuses on group composition in terms of gender and group process functions (such as task and maintenance behavior; Bales, 1950). In other words, we presume, consistent with previous studies (Dindia & Allen, 1992; Mulac, Wiemann, Widenmann, & Gibson, 1988), that groups composed of all men and all women will represent extreme positions on gender-related variables, while mixed groups of both men and women will fall between the extremes. Also we presume, consistent with previous studies (Anderson & Blanchard, 1982; Carli, 1989; Eagly & Karau, 1991), that women will behave consistently with maintenance or socioemotional group process roles and men will behave consistently with task-oriented roles. In general, we anticipate that groups composed entirely of men will perform highest in behaviors such as "being independent, masterful, assertive, and competent" and groups composed entirely of women will perform highest in behaviors such as "being friendly, unselfish, concerned with others, and emotionally expressive" (Eagly & Karau, 1991, p. 686).

The study by Savicki et al. (1996) focused on several important considerations in the examination of gender and CMC. First, the researchers found that, contrary to previous expectation (Herring, 1993), women were not only *not* dissatisfied, but actually were the most satisfied with CMC as

long as they were working in female only (FO) groups. Men in male only (MO) groups were least satisfied with their CMC group experience; and evenly mixed male and female (MIX) groups fell between the two single gender groups on the satisfaction measure. Clearly, this result pointed toward the power of group composition as a factor in satisfaction with CMC, rather than simply gender alone.

Second, consistent with previous research (Herschel, 1994), the prediction concerning participation was that there would be no differences in level of participation between genders. Actual results confirmed that there were no differences between the MO, FO, and MIX groups. Both genders under these conditions participated relatively equally. This result differed from patterns of participation noted on Internet discussion groups (Herring, 1993), perhaps because on the Internet participants probably do not know all the members of the group, and are not working to complete a specific task. Therefore, the context of group interaction seems to be an important factor in levels of participation rather than gender alone.

Third, predictions concerning choice of language by men and women followed those indicated by previous research (Herring, 1993). That is, men would talk more about themselves and their ideas related to the group task, and women would attend more to the socioemotional aspects of group process. Results of the Savicki et al. (1996) study indicated that women in FO groups used more self-referent or 'I' language than did men in MO groups. There was no difference in use of group-oriented or 'we' language between groups. Clearly, simply counting the respective pronoun use did not discriminate the communication processes effectively. A finer discrimination in communication variables was needed.

Finally, predictions concerning interpersonal conflict, or 'flaming' as it is called in CMC, were that men would engage in more acrimonious interchanges in order to avoid changing their opinions; and that women would more actively seek to reduce conflict and would be more likely to abandon their opinions. Results confirmed that men in MO groups did use more coarse language and did depart less from their initial ideas than did FO or MIX groups. However, there was no difference in the amount of tension reduction efforts across groups. Although some predictions were supported, the understanding of how men and women deal with conflict in CMC still needs clarification.

Before stating hypotheses for the variables of interest, a brief review of selected studies will be done regarding the task variable examined in the present study. The task variable in the study will be classified according to two different schemes. First, the task will be classified by content. Research (Holland, 1985) found that men show higher preference for science-oriented, outdoor activities than do women. Tasks whose activities emphasizes such

content would be predicted to be more satisfying to men; likewise, given a probable differential learning history with such activities, men might be predicted to perform more accurately if the task contained such content.

A second classification scheme for task variables comes from McGrath (1984) who has proposed a circumplex model that discriminates tasks according to two dimensions: (a) demands for cooperation versus competition and (b) cognitive versus behavioral activities. Since only cognitive products emerge from CMC, the dimension of interest in the present study categorizes tasks according to the need for interpersonal coordination requiring rich nonverbal communication cues. McGrath and his associates (McGrath, 1984; McGrath, Arrow, Gruenfeld, Hollingshead, and O'Connor, 1993; Straus & McGrath, 1994) predict a relationship between task type, the medium used to perform the task, and indicators of group performance and satisfaction. Specifically for the current study, an intellective task, one in which the group must solve a problem that has a verifiably correct answer, would be predicted to lead to higher satisfaction and performance using the CMC medium than would a decision-making task in which the group had to come to agreement about a problem that has no verifiably correct answer. According to McGrath and Hollingshead (1993) the decision-making task would require more information from group members about feelings, values, and attitudes than would an intellective task. These interpersonal cues are more likely to be transmitted via nonverbal and paraverbal cues which are extremely constrained in a text-based CMC medium. Consistent with the general prediction that women concentrate more on the socioemotional level of group processes, one would expect women to be more satisfied in the decision-making task which puts an emphasis on socioemotional cues.

The current research will seek to clarify questions raised by the Savicki et al. (1996) study in several ways. First, the task used in the 1996 study was a 'feminine'-content, decision-making task in which groups were asked to rank the 'moral character' of five individuals engaged in a hypothetical, 'soap opera-like' interaction. This scenario focused on the internal motivations, thoughts, and feelings of the characters. There were no externally verifiably correct answers to the task. This task was chosen in contrast to more 'masculine'-content, intellective tasks typically used in small group research (e.g., the fallout shelter task or the NASA moon crash) which focus on scientific, outdoor, and survival knowledge and have a verifiably correct answer based on experts' knowledge. The feminine task in the 1996 study may have biased the FO and MO responses; therefore, in the current study both types of tasks are used. Measures of participation, group well-being (satisfaction and group development), and performance will be used.

Given the brief review of gender and task variables, the present study proposes several hypotheses.

H1: There will be no differences in participation due to group composition or task.

H2: FO groups will show most satisfaction with group process while engaged in the feminine-content, decision-making task, and MO groups will show most satisfaction with group process while engaged in the masculine-content, intellective task.

H3: FO groups will show higher levels of group development while engaged in the feminine-content, decision-making task; MO groups will show higher levels of group development while engaged in the masculine-content, intellective task.

H4: MO groups will be more accurate (closer to the expert rankings) in the masculine-content, intellective task than will FO groups.

The current research will use more specific content analysis scales that move beyond mere counting of 'I' and 'we' pronouns. An individual may use 'I' in order to disclose information about him/herself, or 'I' can be used in persuasive, opinion statements. Likewise, 'we' pronouns may be used to establish coalition and cohesion with others in the group, or to establish the basis for persuasion (e.g., "We ought to get started on the task"). The content scales will allow a finer analysis of CMC.

H5: Regardless of task, FO groups will use the most self-disclosure and coalition language; regardless of task, MO groups will use the most opinion and fact language.

Also, this study will examine the effects of group composition and task on conflict or 'flaming' in CMC. Predictions follow results from the Savicki et al. (1996) study.

H6: Regardless of task, FO groups will use less argumentative and coarse language (conflict) than will MO groups.

H7: Regardless of task, MO groups will change their initial ranking less than will FO groups.

As McGrath and his associates indicate (McGrath et al., 1993), research on group activity using CMC contains many variables that must be clarified prior to developing a clear understanding of this phenomenon. In the current study, subjects who had little or no previous experience with CMC, in same or mixed gender groups of 4–6 members, used asynchronous CMC over a 4-week period while working on different tasks. It is hoped that such clarity of context and variables will help focus the findings and the degree to which the findings can be generalized.

METHODS

Subjects

Subjects were 72 undergraduate psychology students. Equal numbers of men and women were randomly assigned to three group composition conditions (MO, FO, and MIX) and two task type conditions (feminine-content, decision-making and masculine-content, intellective). Because the groups were newly formed by random assignment with no previous history of interaction, group variation on the dependent variables is assumed to be related to the independent variables of gender composition and task type; with variation within specific groups offset through the randomization procedure. The groups were not told of their composition differences. Assigned group size was 6 members. A total of 10 subjects never sent an e-mail message beyond the training session and, therefore, were dropped from further analysis making group size range from 4 to 6 members.

Instruments and Measures

Participation. Participation was measured by counting the number of messages sent by each subject and the number of words written in each e-mail message during the group activity.

Satisfaction scale. A six-item locally constructed questionnaire focused on satisfaction with group process and communication aspects of the computer-mediated experience using a 7-point Likert scale. An alpha of .78 in the Savicki et al. (1996) study and an alpha of .76 in the current study indicated satisfactory internal consistency.

Group development. The Team Development Scale (Kormanski, 1990) measured amount of group development reported by members of each group over 10 items, each using a 10-point Likert scale. Kormanski (1990) reports alphas from .92 to .96; an alpha of .92 was reached in the current study.

Performance accuracy. Groups performing the masculine-content, verifiable-answer task were compared with the externally verifiable expert ranking to determine discrepancy from the expert standard.

Measures of choice of language. The manner in which subjects communicated in group activities was measured using four content analysis scales developed by ProjectH (Rafaeli & Sudweeks, 1993), an international group studying CMC. The FirstPerson scale measured verbal self-disclosure, a statement by the author of the message about the author of the message: 0 = yes, 1 = no.

The Opinion scale measured statements of the personal opinion of the message author; it had to indicate the first person directly or indirectly. This scale was adapted into a 2-point scale: 0 = no opinion was present, 1 = opinion was present. The Fact scale measured statement of fact, whether or not the fact was correct, without reference to the first person: 0 = no statement of fact, 1 = one or more statements of fact. The Coalition1 scale measured the degree of agreement with another person or statement previously appearing in the group discussion. This scale was adapted so that only response to previous communicators and its intensity was scaled regardless of its direction; 0 = no reference to another person's message, 1 = mild response to another person on the list, 2 = strong response to another person on the list. Two independent observers rated a subset of the messages to assess reliability of coding. Percent of interobserver agreement [agreements/(agreements + disagreements)] for these scales was 70% for FirstPerson, 76.7% for Opinion, 70% for Fact, and 76.7% for Coalition1.

Conflict. The level of conflict in groups was measured using two content analysis flaming scales developed by ProjectH (Rafaeli & Sudweeks, 1993). The Flame 1 scale measured six levels of argumentativeness of a message from "positive, neutral or no opinion" to "hostile: profanity, tirades, to the point of ignoring original issue". The Flame 2 scale measured 5 levels of the use of coarse or abusive language in a message from "no abusive language" to "abusive language about content and persons in and out of the group". The percentage of interobserver agreement [agreements/(agreements + disagreements)] for these scales was 73% for Flame 1, 100% for Flame 2.

Subject–group task discrepancy. The difference between individual pregroup rankings in the task activity and the postgroup group consensus rankings on the task were computed for each subject.

Procedures

Subjects were trained to use e-mail as the medium for their group activity. All subjects participated in a 1-hr, hands-on, large class, laboratory-based training session in which they learned how to use the Pegasus Mail software (Harris, 1990) to read, send, and reply to electronic messages. Group distribution lists were centrally established on the local area network so that subjects could send and reply to the group list, thus assuring that all group messages were sent to all group members. No group members were identified physically to any other members; all group communication was via e-mail.

After learning to use e-mail, subjects were individually shown their randomly assigned group task and asked to complete their own individual

ranking prior to discussion with their groups. The two group tasks used in this study were selected to match the masculine versus feminine content and the task types identified by McGrath (1984). The feminine-content, decision-making task with no verifiable answer was the 'Lovers' scenario in which five fictional characters (two male, two female, and one of indeterminate gender) were confronted with various moral dilemmas (Savicki et al., 1996). Descriptions were given of their resolutions of the dilemmas. Each character was then ranked (1–5) on the basis of the morality of their actions. There was no independently verifiable right answer. The masculine-content, intellective choice task with a verifiable answer was the commonly used fallout shelter task in which hypothetical threat of a nuclear war forces a group of people to select and rank-order the most important objects to take into a fallout shelter (Johnson & Johnson, 1994, pp. 302–303). They were to rank order the objects (1–5) on the basis of their value to the survival of the group. This exercise has a correct answer developed by experts from the Department of Defense Civil Preparedness Agency. Although the original fallout shelter task included 15 items, only the first 5 were used in this study in order to make the two tasks of equal complexity. No ties were allowed in rankings for either task.

The only structure provided for group interaction was that subjects were prohibited from discussing the assigned task in any other medium, and that they were required to check their electronic mailboxes at least two times per week. Subjects had access to e-mail from any of over 190 networked campus computers which were available 7 days a week for up to 16 hr per day. During the training session it was suggested that they introduce themselves to other group members electronically and that groups such as these accomplished their task by sharing their perceptions and discussing issues in order to arrive at a group ranking.

After 4 weeks of group activity using CMC (during which there was one major network crash), the groups produced a ranking for the assigned task. Each individual then responded to the Satisfaction and Group Development scales.

RESULTS AND DISCUSSION

Generally, group composition was related to the measured group processes in significant ways; however, task differences did not show as strong a relationship. Over the 4-week group interaction period, subjects sent 408 e-mail messages. The following analysis of results sometimes uses individual subjects as the sample, and sometimes uses individual messages as the

sample. In either case, the sample was divided into three types of group composition and two types of tasks.

In the following section, the results will be connected to the research hypotheses. An additional preliminary analysis will be described which may shed some light on the effects of combinations of communication variables to predict levels of group development.

Participation

Two measures of participation were examined: number of messages sent and number of words sent. For the number of messages sent, a factorial analysis of variance indicated that there were no differences in participation for either group composition or task type. However, for the number of words sent, analysis of variance indicated a significant main effect for group composition ($F(2) = 12.523$, $p < .0001$) but not for task type. FO groups sent significantly more words than either MIX or MO groups who were not different from each other. Therefore, although FO groups did not send more discrete messages, they did use more words per message than did either MO or MIX groups (Table 1).

When participation was measured by number of messages sent, Hypothesis 1 was supported since no differences appeared. However, when participation as measured using number of words written, the hypothesis was not supported. In the Internet context, male groups send longer messages; however, on the Internet participants probably do not know all the members of the group and are not working to complete a specific task. It seems that the context of this study, small groups in which all members contributed to a well-defined task, allowed women to participate more fully. Research on conditions necessary to develop an effective group (Johnson & Johnson, 1994) indicates the need for clear goals and relatively equal participation. Thus, it is not surprising that the context in this study would lead to better participation from individuals predicted to have a greater sensitivity to group process.

Table 1. Group Means and Standard Deviations for Number of Words

	Group Composition								
	Female Only			Mixed			Male Only		
Task Type	*M*	*SD*	*n*	*M*	*SD*	*n*	*M*	*SD*	*n*
Lovers	98.21	54.59	72	84.43	57.73	67	77.37	77.61	84
Fallout shelter	124.82	85.89	71	82.87	67.06	60	66.11	50.26	54

Satisfaction Related to Task Type

A factorial analysis of variance indicated significant differences for group composition in the satisfaction measure ($F(2) = 7.629$, $p < .02$, but not in task type (Table 2). FO groups were significantly more satisfied than either MIX or MO groups who were not different from each other.

The hypothesis that satisfaction would be related to task type was not supported since satisfaction with group process was related only to group composition. FO groups, regardless of task, were most satisfied. Neither the content of the task (masculine or feminine) nor the process of the task (intellective vs. decision-making) lead to hypothesized differences. Task type seems unrelated to satisfaction.

Group Development Related to Task Type

A factorial analysis of variance indicated significant differences for group composition in the group development measure ($F(2) = 16.592$, $p < .0001$, but not for task type (Table 3). FO groups scored themselves significantly higher in group development than either MIX or MO groups, who were not different from each other.

The hypothesis that group development was related to task type was not supported since FO groups indicated highest levels of group development, regardless of task type. Task type seems unrelated to group development.

Accuracy of Group Performance

A factorial analysis of groups engaged in the masculine-content, intellective task showed a marginally significant difference for group accuracy of ranking

Table 2. Group Means and Standard Deviations for Satisfaction With Group Process

	Group Composition								
	Female Only			Mixed			Male Only		
Task Type	*M*	*SD*	*n*	*M*	*SD*	*n*	*M*	*SD*	*n*
Lovers	38.00	4.27	10	33.00	5.29	10	34.13	3.36	8
Fallout shelter	38.63	3.96	8	32.88	5.17	8	33.25	3.92	8

Table 3. Group Means and Standard Deviations for Group Development

	Group Composition								
	Female Only			Mixed			Male Only		
Task Type	*M*	*SD*	*n*	*M*	*SD*	*n*	*M*	*SD*	*n*
Lovers	92.90	8.92	10	73.80	10.87	10	76.25	9.51	8
Fallout shelter	90.13	8.98	8	74.25	12.12	8	74.13	9.52	8

as compared with the expert rankings, $F(2) = 2.774$, $p = .08$. MO groups showed the most discrepancy from the expert rankings and FO groups the least discrepancy, with MIX groups falling between. This result, although not reaching the .05 level of significance, is interesting because it is exactly the reverse of the hypothesized direction. Caution must be exercised in considering this result because of the marginal significance and the halving of subjects used in the analysis.

In summary, group composition variables seem to show significant relationships to group well-being measures (satisfaction and group development), a marginal relationship to group performance measure (accuracy) and mixed relationships to participation measures (number of messages and word count). Task type was not related to any of these variables.

Choice of Language Related to Group Composition

Chi-square analysis indicated significant relationships between use of FirstPerson both with group composition, $\chi^2(2) = 14.043$, $p < .0001$, and with task type, $\chi^2(1) = 5.033$, $p < .03$ (Table 4). Examination of Table 4 indicates that FO groups had the largest percentage of messages containing the FirstPerson (65%), followed by MIX (54%) with MO groups using the least FirstPerson (43%). Differences in use of FirstPerson by task type show that the fallout shelter task had higher levels (60%) than did the lovers task (49%).

Chi-square analysis indicated a significant relationship between group composition and the use of Opinion, $\chi^2(2) = 13.089$, $p < .002$, but not task type (Table 4). Examination of Table 4 indicates that FO groups had the

Table 4. Observed Frequencies for Choice of Language Content Scales in Messages by Group Composition and by Task Type

	Group Composition			Task Type		
Content Analysis Scale	Female Only	Mixed	Male Only	Lovers	Fallout Shelter	Total
FirstPerson						
No first person	50	59	79	114	74	188
First person	93	68	59	109	111	120
Total	143	127	138	223	185	408
Opinion						
No opinion	3	17	9	15	14	29
Opinion	140	110	129	208	171	379
Total	143	127	138	223	185	408
Coalition1						
No coalition	89	94	112	164	131	295
Mild coalition	40	19	18	47	30	77
Strong coalition	14	14	8	12	24	36
Total	143	127	138	223	185	408

largest percentage of messages containing Opinion (98%), followed by MO (93%) with MIX groups using the least Opinion (87%).

No significant differences appeared in the use of Fact in messages for either group composition or task type variables.

Chi-square analysis indicated significant relationships between the use of Coalition1 both with group composition, $\chi^2(4) = 15.621$, $p < .004$, and with task type, $\chi^2(2) = 7.975$, $p < .02$ (Table 4). Examination of Table 4 indicates that FO groups had the largest percentage of messages containing mild or intense reactions to another person in the group (38%), followed by MIX (26%) with MO groups showing the least reaction to others in their group (19%). Groups engaged in the fallout shelter task had the largest percentage of messages containing mild or intense reactions to another person in the group (29%), while lovers task groups had a lower number (26%). The difference between task conditions seems to be accounted for by the differences in the larger number of more intense responses in the fallout shelter task (13% vs. 5%) and the larger numbers of mild responses in the lovers task (21% vs. 16%).

These results lead to a mixed reaction to the hypothesis that choice of language is related to group composition. First, in relation to Hypothesis 5, FO groups showed significantly larger numbers of messages containing FirstPerson (self-disclosure), as predicted. FO groups also used the most Coalition1 language in which they overtly responded to previous messages in the group, as predicted. However, they also showed significantly larger numbers of messages with Opinion, the opposite of the hypothesized direction. MO groups did not use more Fact language since there were no group differences. Further clarification of the relation of gender and communication patterns is needed. As in the previous Savicki et al. (1996) study, FO groups used significantly more 'I' language ('I', 'me', 'my') than did either MO or MIX groups, $F(2) = 18.138$, $p < .0001$. 'I' statements may be serving some function other than advocating the sender's individual position or opinion. Simple gender predictions and simple group composition predictions need to be more discriminating.

Both FirstPerson and Coalition1 measures were significantly more frequent in the masculine-oriented, intellective, fallout shelter task. This result is opposite of the expected direction, since an intellective task theoretically requires less personal contact from participants than does a decision-making task. Although not predicted, another interesting task type difference did emerge. More group-oriented, 'we', language was used in the fallout shelter task, $F(1) = 7.728$, $p < .01$. Contrary to expectations, this masculine-content, intellective task may have drawn more group-oriented language because the assignment focused on hypothetical group survival.

The lovers task, in contrast, may have drawn less 'we' language since the criterion for acceptance of one's ideas was not the 'survival' of the group, but rather the force and logic of the opinion in the absence of an externally verifiable criterion. Task type seems more complicated than initial predictions indicated. Clearly, even for the masculine-oriented, intellective task there was a strong socioemotional component indicated by choice of language.

Conflict Related to Group Composition

Chi-square analysis indicated a significant relationship for group composition and use of Flame 1 (argumentativeness), $\chi^2(6) = 12.922$, $p < .05$, but not task type (Table 5). Examination of Table 5 indicates that MO groups had the largest percentage (6%) of messages containing tension (attacking an opposing argument), followed by MIX (2%) with FO groups using no tension (0%).

Chi-square analysis indicated a significant relationship with task type and the use of Flame 2 (coarse and abusive language), $\chi^2(3) = 24.214$, $p < .0001$, but not with group composition (Table 5). Examination of Table 5 indicates that lovers task groups had the largest percentage (15%) of messages containing coarse or abusive language (mostly toward content not toward others in the group), with fallout shelter groups showing less (4%). This level of coarse language may be an artifact of the moral dilemmas posed by the lovers hypothetical scenario which invited unsavory descriptions of some of the characters based on their actions in the scenario.

Table 5. Observed Frequencies for Conflict Content Scales in Messages by Group Composition and by Task Type

	Group Composition			Task Type		
Content Analysis Flaming Scale	Female Only	Mixed	Male Only	Lovers	Fallout Shelter	Total
Flame 1, Argumentativeness						
Neutral or friendly	126	103	114	193	150	343
Diverging	1	0	0	0	1	1
Disagreeing	16	21	16	25	53	53
Tension	0	3	8	5	6	11
Total	143	127	138	223	185	408
Flame 2, Use of Coarse or Abusive Language						
None	131	116	121	190	178	368
About content only	12	9	12	31	2	33
About a group member	0	0	0	0	0	0
About generalized others	0	1	5	2	4	6
Mixed	0	1	0	0	1	1
Total	143	127	138	223	185	408

The support for the predicted relationship between conflict and group composition was mixed. MO groups did use more Flame 1, argumentativeness; however, for Flame 2, use of coarse and abusive language, no group composition differences appeared. The difference in coarse and abusive language used in the lovers task was opposite of the expected direction, since an intellective task has a lower need for collaboration and cooperation; thus, allowing more individually extreme language.

Individual Versus Group Differences

A factorial analysis of variance showed no significant differences for either group composition or task type in terms of the change that individuals made between their initial rankings and the final rankings the groups made.

Post Hoc Analysis: Choice of Language and Group Development

Group development has many linkages to group performance and well-being (Kormanski & Mozenter, 1987). It would be interesting to examine possible connections between group development and CMC patterns. Given the connectedness of group composition to both group development and to choice of language as indicated (even if not always in the hypothesized direction), a post hoc analysis was conducted to test whether a specific pattern of choice of language might be predictive of levels of group development. In simpler terms, how are group members (especially in FO groups) communicating that may lead to more advanced group development?

In order to answer this question, a hierarchical multiple regression was done on all subjects regardless of group composition and task type conditions. The first step in this two-step multiple regression controlled for varying levels of participation. Table 6 shows marginal significance for the relationship of group development to both number of messages and number of words, $R^2 = .11$, $F(2) = 3.067$, $p = .06$. In step two, all of the content analysis measures were added. These measures were summed over messages for each subject, therefore, they appear as cumulative interval scores which can be related directly with the group development score. Table 6 shows that choice of language content scores add significantly to the relationship with group development and participation alone, $R^2 = .36$, $F(8) = 3.017$, $p < .01$). The combination of choice of language content scores that contributed significantly were Coalition1, standardized $\beta = .423$, $p < .004$, and Flame 1, standardized $\beta = -.737$, $p < .03$. In summary, higher levels of group development were related to addressing messages to specific others in the group and avoiding argumentativeness.

Table 6. Hierarchical Multiple Regression of Group Development With Participation, Choice of Language, and Conflict Measures

Variable	Standard Coefficient	*t* Value	*p* Value
	Step 1, R^2 = .11, $F(2)$ = 3.067, p = .056		
Word count	.446	2.201	.033
Number of messages	−.486	−2.390	.020
	Step 2, R^2 = .36, $F(8)$ = 3.017, p < .010		
FirstPerson	−.277	−1.341	*ns*
Opinion	−.301	−0.499	*ns*
Fact	−.188	−1.001	*ns*
Coalition	.423	3.049	<.004
Flame 1	−.737	−2.299	<.030
Flame 2	.190	0.429	*ns*

CONCLUSIONS

In conclusion, several issues and concerns emerge related to gender and CMC. First, it remains clear that gender is a factor that must be considered in CMC. However, when men and/or women use CMC in groups, group composition seems to be a powerful mediator of gender effects. Also, as results here show, the direction of the effects are not always as predictable as previous research might indicate. Further research needs to be focused on the contexts in which genders interact. Contrasting results between small group interaction and Internet discussion groups highlight the effects of context on gender and CMC.

Second, the lack of task effects points to the need for a finer distinction of tasks and task demands. Results of the current study are somewhat similar to those of Straus and McGrath (1994) who found that although some task results matched predictions, others did not. On the one hand, a distinction between intellective and decision-making tasks may be too fine to call forth the predicted results reliably; on the other hand, other task characteristics may need to be accounted for. For example, in the current study, the lovers task invites name-calling of characters. Such harsh evaluation used toward the task (not toward others in the group) may still contribute to a 'defensive climate' (Gibb, 1961) in which group members refrain from self-disclosure or even from personal opinions. Different tasks may call forth responses that conflict with group effectiveness and well-being.

Third, Herring (1994) indicates that CMC groups can be 'gendered' in the sense that their norm of interaction more closely follows female or male language choices. In her research, Herring identified contrasting values espoused by men and women for CMC. "Women place a high value on consideration for the wants and needs of others", "Men, in contrast, assign a greater value to freedom from censorship, forthright and open expression, and agonistic debate as a means to advance the pursuit of knowledge"

(Herring, 1994, p. 7). The inverse relationship between Flame 1 and group development in the current study indicates that, in the CMC context, group well-being must take conflict into consideration. Absence of attention to the socioemotional level of group activity is related to low satisfaction and group development.

Finally, it would be useful to define styles of communicating that enhance group development in the CMC context. In an emerging work group, norms might be set to mimic communication styles that enhance group development in CMC. FO groups were able to overcome the inherently sparse nonverbal cues available via text-based CMC. Instead of using compensatory devices such as emoticons (smileys), they systematically responded to others in their group and avoided flaming. The extensive use of 'I' statements found in an earlier study (Savicki et al., 1996) and replicated in this study seem to encompass both self-disclosure and opinion. In addition, other, uncoded, 'I' statements were liberally sprinkled through the messages. These 'I' statements may function to make the sender more 'present' in the text-based medium and to avoid the 'collective monologue' created by a succession of fact-oriented statements that do not reveal much about the sender. Indeed, the MO groups showed all the features of the collective monologue described by Hewes (1986) in which members were talking at each other rather than to each other. In contrast, the FO groups emphasized a connecting device that Hewes calls 'acknowledgments' (Hewes, 1986, p. 282). In a collective monologue, the acknowledgments are 'vacuous'; in a true discussion they are substantive. The strong relationship between group development and Coalition1 indicates that the FO groups' acknowledgments were substantive. It is possible that these communication styles can be taught.

One potential application for the results of this study is training groups in the purposeful use of those group development enhancing communication patterns in order to overcome the 'facts only,' collective monologue tendency of some groups. Such training would need to take into account the context of group activity. Research is needed to determine if large aggregations of discussion group members (as on the Internet) can form the kind of 'groupness' that developed in the small, specific task-oriented groups in this study.

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